



Jiadong Lu

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Availability: Available full-time for a credit-bearing internship; immediate start; 3–6 months.

Languages: Proficient in Chinese and English. **Pass status:** Singapore Student's Pass holder.

Education

National University of Singapore (QS Top 10)

Aug 2025 – Jun 2027 (expected)

Master of Science in Biomedical Informatics

Zhuhai College of Jilin University (Rank: 1/235)

Oct 2020 – Jun 2024

Bachelor's Degree in Data Science and Big Data Technology

Technical Skills

- **Health-data and AI workflows:** Conversation-data curation and quality checks; pathology image analysis; patient-context extraction; retrieval-augmented generation; longitudinal follow-up; model evaluation and failure analysis.
- **Model training and deployment:** LoRA and full-parameter SFT, DPO, multi-turn RL and distillation; vLLM; multi-node, multi-GPU training, GPU memory optimisation and training diagnostics.
- **Programming and tools:** Proficient in Python; LangChain, LangGraph, AutoGen; Linux, Git, Weights & Biases; collaborative AI-assisted development.

Professional Experience

Kriston.AI

May 2026 – Oct 2026

LLM Post-Training Engineer (Intern)

AI Algorithms Team 1 | Paediatric Enquiry and Sales Assistant

Developed a specialised model for multi-turn paediatric sales and customer-service conversations.

- **Data preparation and deployment.** Combined user-simulated conversations with real production logs, applied a shared quality rubric and balanced topics, dialogue lengths and out-of-distribution examples. Fine-tuned **Qwen3.5-27B with LoRA SFT** and deployed it after benchmark iterations and simulation-based acceptance testing.
- **Evaluation and continuous improvement.** Analysed production traces, repaired failure cases and fed improvements back into SFT and DPO, reducing the **failure rate from 25% to 5%** on a dynamic evaluation set of **1,000 real business examples**. Trained topic-specific teacher models and validated multi-teacher on-policy distillation into Qwen3.5-9B; achieved the best performance among evaluated models across all three tiers of the evaluation framework.

AI Algorithms Team 2 | Unified Model across Medical Specialties

Contributed to a shared model for healthcare sales-lead conversion to replace specialty-specific models and proprietary APIs.

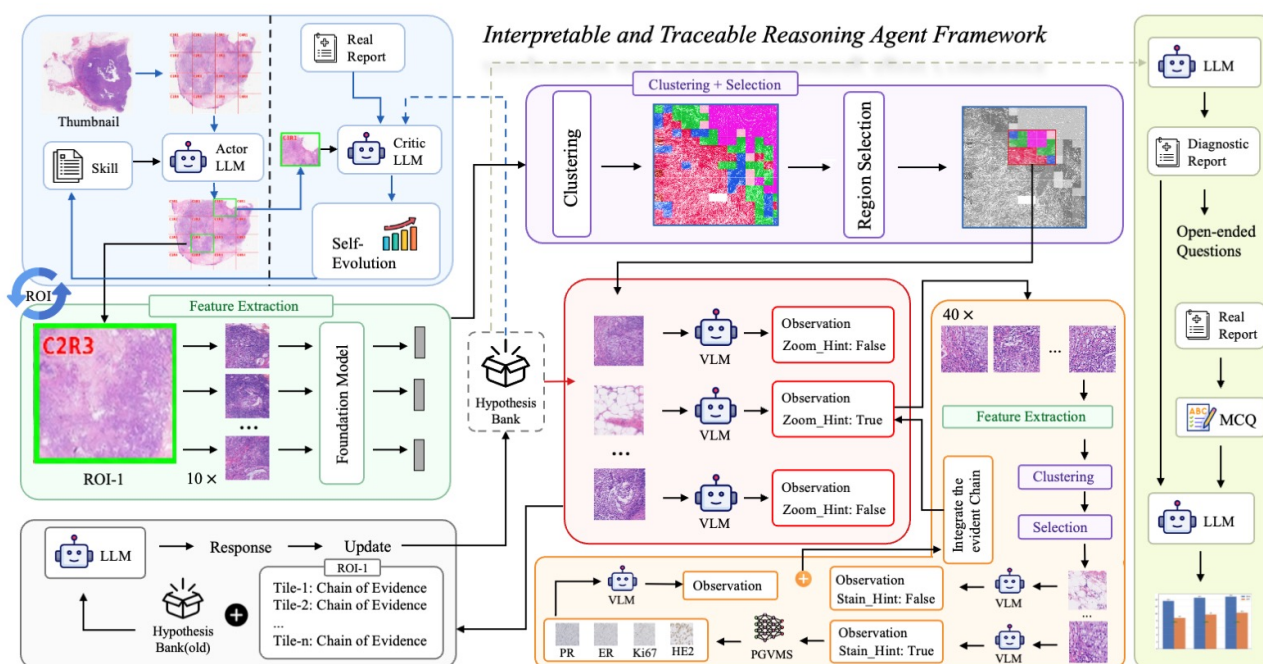
Cold-Start Model and Production-Data Feedback

- Contributed to **two-stage full-parameter SFT of Qwen3-8B-Base / Qwen3-8B-CPT**. Stage 1 used approximately **460,000 production logs** to build multi-turn training data for empathetic, goal-directed consultation and sales-conversion workflows. Stage 2 targeted final-turn responses in difficult examples and production failures.
- Built the **data feedback loop** from log ingestion and failure analysis through response repair, quality validation, retraining and regression evaluation, adapting the model to real user distributions and interaction patterns.

Agentic RL | User Simulation and Multi-Turn Training

- **Interaction modelling.** Recreated production agent workflows and built a user simulator from real conversations and behavioural distributions, modelling user state, feedback and stopping behaviour to improve training-environment fidelity.
- **Training and validation.** Combined trajectory-level business quality and final conversion with turn-level user-state changes; used return-to-go to calculate per-turn advantages. Built a multi-turn **DAPO** training pipeline and refined it through trajectory audits and evaluation. Improved internal **Excellent-Bench performance by 55%**, with other evaluated capabilities maintained; the deployed model handles **100,000+ calls per day**.

- Evidence-grounded pathology agent.** Built a **breast-cancer whole-slide image (WSI)** workflow linking region selection, observation, hypothesis updates and further evidence gathering. Combined foundation-model feature clustering with a vision-language model for multi-magnification inspection, tracing diagnostic conclusions to image evidence.
- Region-selection refinement.** Designed an Actor–Critic mechanism to evaluate region selection offline against pathology reports and capture feedback as reusable agent skills for iterative exploration.
- Expert-reviewed evaluation.** Built a benchmark of **1,210 WSIs and 3,712 multiple-choice questions** from reports annotated and reviewed by senior pathologists. Outperformed the evaluated frontier multimodal LLM baselines and baselines on all public benchmarks tested.



- Conversational support for chronic care.** Co-developed a **chronic-care AI agent delivered through WeCom (WeChat's enterprise platform)**, from requirements modelling to implementation. Built query rewriting, parallel multi-intent detection, hybrid retrieval and multi-stage generation workflows for **diet, exercise, weight and blood-glucose management**.
- Patient-context memory.** Fine-tuned a domain memory model on real chronic-care conversations with **LoRA SFT** to resolve omitted context and implicit references. Deployed it with **vLLM** and integrated it into agent memory, raising **key-information extraction accuracy from 80% to 95%**.
- Proactive, longitudinal follow-up.** Orchestrated follow-up SOPs as calendar tasks with trigger times, executable skills and tracked status. A heartbeat mechanism drove information collection, health-indicator analysis and proactive reminders, writing results and patient state back to persistent task memory.
- Coordinated care workflows.** Contributed to decomposing a single-agent workflow into medical-expertise, nursing and data-analysis agents for guideline interpretation, follow-up, patient-data queries and indicator calculations, reducing coupling and improving maintainability and interaction stability. A related manuscript is **under review at npj Digital Medicine** (5-year IF: 19.7, 2025).

- Enhanced Word-Unit Broad Learning System With Sememes.* IEEE Access, **IF 4.2. Second author.**
- Typhoon Track Prediction Based on TimeForce CNN-LSTM Hybrid Model.* Springer. **First author.**